

A Sustainable Approach to Ex Situ Seed Conservation through Citizen Participation and Distributed Storage

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Abstract—More than 95% of the crop genetic erosion articles analyzed in [1] reported changes in diversity, with nearly 80% providing evidence of loss. The lack of diversity presents a severe risk to the sustainability of global food systems. Without seed diversity, it is difficult for plants to adapt to pests, diseases, and changing climate conditions. Genebanks, such as the Svalbard Global Seed Vault, are valuable initiatives to preserve seed diversity in a single secure and safe place. However, according to our analysis of the data available in the Seed Portal, the redundancy for some species might be limited, posing a potential threat to their future availability. Interestingly, the conditions to properly store seeds in genebanks, are the ones available in the freezers of our homes. This paper lays out a vision for Collective Seed Storage relying on a peer-to-peer infrastructure of domestic freezers to increase the overall availability of seeds. We present a Proof-of-Concept focused on monitoring the proper seed storage conditions and incentivizing user participation through a Blockchain lottery. The PoC proves the feasibility of the proposed approach and demonstrates that our vision can contribute to several of the 17 United Nations Sustainable Development Goals.

I. INTRODUCTION

There are over 50,000 edible plants worldwide, but just 15 provide 90% of the world's food energy intake¹. We have to put all our efforts to preserve those fundamental energy assets.

The need for seed diversity. A key objective in preserving these plants is safeguarding seed diversity, namely the variety of plant species, cultivars, and genetic resources within crops. This diversity represents a fundamental asset for the future of food systems. As climate change intensifies and new pests and diseases emerge, specific genes enabling crops to tolerate heat, drought, pathogens, and other stresses become essential for agricultural resilience and adaptation.

The erosion of seed diversity not only limits the ability of agriculture to respond to environmental changes but also threatens nutritional value, flavor diversity, and the cultural heritage associated with traditional crops. Ensuring access to a broad range of crop varieties is therefore critical for enabling future generations to adapt agriculture to evolving environmental conditions and societal needs.

Conservation efforts should also include Crop Wild Relatives (CWRs), which provide valuable genetic resources

that may be absent from current cultivated varieties and can contribute to improving crop productivity, resilience, and quality. However, plant genetic diversity is declining at an unprecedented rate: more than 95% of crop genetic erosion studies analyzed in [1] reported changes in diversity, with nearly 80% documenting evidence of loss. In many regions, this erosion has already reduced the range of options available to farmers.

Efforts to preserve seed diversity. Preservation of seed diversity is considered a common fundamental goal among the countries organized in the Multilateral System of Access and Benefit Sharing (MLS [2]), part of the International Treaty on Plant Genetic Resources for Food and Agriculture (PGRFA). MLS allows member states to share and exchange plant genetic resources for research, breeding, conservation, and training. Recipients sign a Standard Material Transfer Agreement (SMTA) [3] that specifies conditions for use and benefit sharing. MLS includes 64 of the world's most important crops that together represent 80% of all human consumption derived from plants.

The main strategies to preserve seed diversity are conservation *in situ* [4] and *ex situ*. *In situ* conservation relies on the conservation of seeds and plants in their natural or traditional environment, such as farms where traditional varieties (landraces) are maintained or in the wild where CWR grow. Participatory seed breeding and community seed systems engage local farmers in storing, conserving, sharing, and improving diverse crop local varieties adapted to local conditions. Although *in situ* conservation has the merit of actively participating local communities and farmers, thus contributing to the conservation of not only the seeds, but also the traditions and skills to take care of them, the access to the material and the documentation is fragmented and difficult to handle. These problems have led to the development of several more structured *ex situ* conservation initiatives, where seeds are preserved outside their natural habitat in the so-called *seed banks*.

Seed banks. A seed bank is a gene bank, that stores seeds to preserve genetic diversity, namely the genes that make each plant variety unique. It ensures that this genetic heritage is safely conserved and available for people to use.

¹<https://education.nationalgeographic.org/resource/food-staple/>

Criteria	In Situ Conservation	Ex Situ Conservation
Location	Natural environment / farms	Seed banks, seed vaults, labs
Evolution	Plants continue to adapt naturally.	Static; no natural evolution
Risk of Loss	Higher (e.g., land-use change, climate)	Lower (secure and controlled conditions)
Cultural Link	Strong; maintains traditional knowledge	Weak or absent
Access & Sharing	Local and informal	Global access via formal mechanisms (e.g., SMTA)
Documentation	Often limited or informal	Detailed and standardized
Cost	Lower short-term cost	Higher cost (infrastructure and maintenance)
Best Use	Wild relatives, landraces	Major crops, backup, and global breeding

TABLE I
COMPARISON BETWEEN IN SITU AND EX SITU SEED CONSERVATION METHODS.

The Svalbard Global Seed Vault in Norway (SGSV)², is unanimously considered a global backup facility for seed banks worldwide [5]. Its purpose is to provide long-term backing for genebank collections to secure the foundation of our future food supply against threats such as mismanagement, accidents, equipment failures, funding cuts, war, sabotage, disease, and natural disasters. In 2019, seeds were returned from Svalbard to restock an international genebank destroyed by the war in Syria.

Contrary to the Millennium Seed Bank Partnership (MSB [6]), SGSV is not open to researchers and the depositors are the only ones who can withdraw their own seeds. The focus of SGSV is to preserve as much diversity as possible within crop species and their wild relatives (intraspecific diversity).

Seed banks must ensure that their holdings remain alive. They periodically check the viability of the samples, namely their ability to grow into productive plants and produce fresh material when needed. However, the fundamental strategy to preserve such diversity is the availability of good backup systems, namely other seed banks where duplicate samples are delivered for safe keeping.

Seed preservation pipeline The standards for seed preservation [7]–[9] recommend implementing a comprehensive preservation pipeline. The underlying principle is that safe storage alone is insufficient; it is equally essential to establish the conditions and maintain the knowledge necessary to reliably propagate seeds over time. In this paper, however, we focus exclusively on the *drying and storage* step of the pipeline, motivated by the open problem discussed in the next section. In Section V, we explain why, despite this limitation, we consider this step a fundamental contribution toward the eventual implementation of the full preservation pipeline in a fully distributed setting, encouraging citizen participation.

Open problem The SGSV is unanimously considered the backup seed facility that provides the highest standards of safety and security. The 2023 SGSV Annual Progress Report [10] reveals that, 1267127 samples deposited by more than 100 genebanks/institutes and representing more than 6,000 plant species are stored in the facility. The SGSV now safeguards seeds of more than 250,000 types of wheat, 160,000 types of rice and 46,000 types of maize, the fundamental sources of food energy intake.

Our analysis of the data from the SGSV Seed Portal³, shows that the vast majority of species have a single depositor. In other words, the data show that the samples are available at SGSV and at the depositor only. As an example, *Oryza sativa* (i.e. the Asian cultivated rice) is the most common rice cultivated as a cereal. The result of the query to the Seed Portal with the keyword *Species: Oryza sativa* is 29 depositors, 171193 accessions and 136 countries of collection. The same query with the keyword *Species: Hygroryza aristata*, gives 1 depositor (i.e. the Rice Research Institute), 4 accessions and 1 country of collection as also confirmed by the Genesys online platform. Rice is currently by far more “important” than Hygroryza for food supply, however, the Hygroryza is a CWR of rice [11] and, as already observed, it might contribute to the development of more resilient new rice varieties.

The low level of redundancy in the availability of seeds samples backups for the majority of the species, might have serious consequences; even in the best genebanks, things can occasionally go disastrously wrong. The question we aim to investigate in this paper is whether it is possible to further increase the overall availability of seeds by a robust and resilient backup infrastructure capable of preserving seed diversity with limited investments.

Contributions of the paper. A comprehensive strategy for seed preservation must address all components of the conservation pipeline and should actively encourage citizen participation to raise awareness and engagement. Nonetheless, having a robust and resilient backup remains a fundamental element to support the sustainability of that strategy. In this paper, we focus specifically on this essential, though narrowly defined, objective.

Interestingly, the temperature conditions in domestic freezers can meet the Food and Agriculture Organization (FAO) standard for long-term seed conservation [7], [8]. The main idea of this paper is to develop a system inspired by the principles of the shared economy, capable of safely storing the seeds in our domestic freezers, thus contributing to the development of a Collective Seed Storage (CSS) that can increase the availability of seeds and the overall resilience of the world’s crop diversity. To our knowledge, this paper is the first attempt to build a peer-to-peer infrastructure to preserve seed diversity. We first identified a set of requirements, and then we built a proof-of-concept of the CSS that proves the

²<https://www.seedvault.no/>

³<https://seedvault.nordgen.org/Search>

technical feasibility of two fundamental tasks, namely the IoT monitoring activity of seed conservation conditions [12] in domestic freezers and the interaction of the IoT node with a Blockchain smart contract to autonomously manage a lottery to incentivize user participation. The development of this proof of concept (PoC) enabled the identification of key challenges that warrant further investigation in future work. Moreover, it demonstrates that the CSS can contribute to several of the 17 United Nations Sustainable Development Goals (SDGs)⁴, as discussed in Section VI.

II. REQUIREMENTS

In this section we describe the main requirements that drive the design of the CSS proof-of-concept.

- R1** Storing conditions of seeds in domestic freezers must comply with the FAO standards [7], [8]. In this paper, we focus on long-term storage of orthodox seeds. Specifically, we assume that after drying, samples meant for long-term storage are packaged under controlled conditions are ideally stored at $-18^{\circ}\text{C} \pm 3$ percent and relative humidity of 15 ± 3 percent (Standard 4.2.3 [7]). Subzero freezer storage is acceptable if ideal conditions are not available [8].

The success of this project relies on the active and numerous participation of users. To encourage participation, the system should:

- R2** Simplify the installation without requiring any specific technical skill or infrastructure. The installation of the system should be user-friendly and based on off-the-shelf equipment. We only assume users have WiFi at home.
- R3** Run for at least 1 year without the need of intervention by the user.
- R4** Support the implementation of suitable incentive mechanisms.

III. PROOF-OF-CONCEPT

The main purpose of this section is to demonstrate the feasibility of the proposed approach by the implementation of a Proof-of-Concept (PoC) of the Collective Seed Storage driven by the above requirements. The PoC will also outline the main technical problems that still need to be efficiently solved to realize a fully fledged solution.

The airtight container storing the seeds is equipped with a humidity and temperature sensor (the DHT22) connected to an IoT sensor node, in our case the ESP32 Devkit V1 (ESP32 in the following), which implements the logic to acquire the sensor readings and to deliver them to the smart contract in charge of dispensing the incentives to the participants.

The ESP32 delivers the data to the smart contract by the on-board WiFi. WiFi access points are nowadays largely available in homes (see **R2**) making this technology appropriate for the project, furthermore, the nature of the phenomenon to be monitored does not require frequent data acquisition; temperature

and humidity in the freezer do not suddenly change. For this reason, the node alternates long periods of *deep sleep* with a short active period where two main tasks are performed, namely data acquisition and communication. In *deep sleep* mode, only the RTC Timer and RTC Memory are active, reducing the consumption to $10\mu\text{A}$. The resulting low duty cycle prolongs the lifetime of the node, and allows us to meet the requirement **R3** as quantified in section III.

The ESP32 is powered by a 3.6V LiFePo4 to avoid unnecessary power cables and simplify the installation. This solution does not require any voltage regulator that would introduce losses. In the future, we plan to design a more efficient custom board without all unnecessary components.

The ESP32 operating temperature range⁵ is from -40 to 125°C , so in principle it can also be placed into the freezer together with the DHT22 temperature and humidity sensor, however, the WiFi connectivity could be severely hindered. In our experiments, when the ESP32 is placed into the freezer, more than 40% of the packets do not reach the server. Furthermore, it is well known that the efficiency of batteries decreases significantly in cold temperatures: at -20°C to -40°C , a LiFePo4 battery may only achieve about 60% to 40% of its rated capacity. Finally, despite being small, the ESP32 steals some further space in the user's freezer. For these reasons, the DHT22 is placed into the freezer and it is connected to the ESP32 placed outside, by three tiny wires.

A sketch of the architecture of the Proof-of-Concept is shown in figure 1. The PoC is built with PlatformIO⁶ and the code is available on https://github.com/andreavitaletti/PlatformIO/tree/main/Projects/web3E_SC.

In the following, we discuss the implementation of the two main tasks performed by the PoC during the active period, namely monitoring the correct storage of seeds and delivering the observed data to the smart contract in charge of incentivizing user participation.

Monitoring the Correct Storage of Seeds. The temperature and humidity conditions at which seeds are stored in the freezer, are monitored by a DHT22 sensor (see **R1**). It features excellent technical specifications also in terms of power consumption: $40\mu\text{A}$ in stand-by-mode 1mA when the temperature/humidity is required.

Humidity variability mainly depends on the initial moisture content of the seeds, the drying and conditioning procedures prior to storage, and the moisture barrier properties of the storage container. For this reason we initially focus only on the temperature conditions. The results of a monitoring campaign conducted on a domestic freezer (BOSCH KGN39X23) set to -18°C are shown in Figure 1. Temperature samples were collected every 5 minutes. After an initial transient period, the measured temperature exhibited significant oscillations. Although these conditions do not meet the ideal storage requirements specified in Standard 4.2.3 [7], they are consistent with the acceptable subzero storage conditions recommended

⁴<https://sdgs.un.org/goals>

⁵https://www.Espressif.com/sites/default/files/documentation/ESP32_datasheet_en.pdf

⁶<https://platformio.org/>

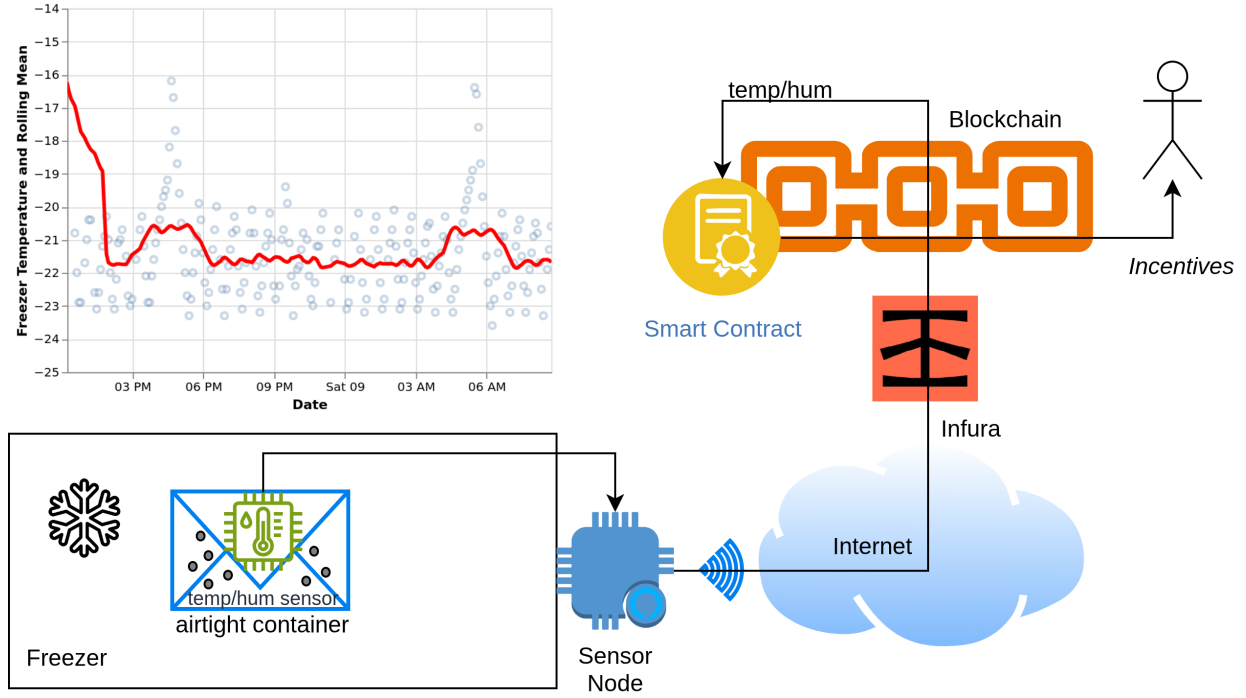


Fig. 1. The architecture of the Proof-of-Concept for the Collective Seed Storage. A temperature and humidity sensor is placed into the freezer and it is connected to an IoT sensor node placed outside. The observed temperature drops immediately after the placement of the sensor into the freezers, then fluctuates noticeably. The node communicates the readings on the conservation conditions to a smart contract in charge of dispensing incentives to encourage users' participation.

in [8] when ideal conditions cannot be achieved. Since the objective of this work is to assess the feasibility of achieving FAO-compliant storage conditions, rather than directly measuring biological outcomes, a dedicated seed viability study is beyond the scope of this paper.

Delivering the observed data to the smart contract.

Once acquired, the data are delivered to a blockchain smart contract that distributes incentives to encourage user participation (see **R4**), following an approach similar to that proposed in DePIN [13]. In our PoC, we propose a blockchain-based lottery that rewards active participation: users are eligible to enter only if they contribute temperature and humidity measurements that remain within an acceptable range, thereby demonstrating that the seeds are being stored under appropriate conditions. A minimalistic version of the smart contract implementing the lottery, along with a set of sample transactions, is publicly available at <https://sepolia.etherscan.io/address/0x3d5983543381dFEd2Dd41d9292Ba86c58C582Be7>. We interact with the smart contract through the Infura API⁷.

Energy Consumption. In this section we report on the experiments to evaluate the energy consumption through an INA219⁸. The ESP32 devkit V1, has an always on red led, with a fix consumption of 10mA. We unsoldered the led and the consumption dropped to less than 4mA, however still pretty far from the nominal 10 μ A in deep sleep. In the following, to

evaluate the potentials of our work, we assume to achieve the nominal current consumption during the sleeping period.

We sampled several active periods and figure 2 shows the current consumption in one of those periods; similar values have been observed for the others. Active periods are made of two tasks, the *init* task where all the necessary components are initialized, and the *com* where the data are read by the DHT and are transmitted to the smart contract. The *init* task lasts about 2400ms (≈ 0.0007 h) at about 50mA, namely ≈ 0.03 mAh, while the *com* task lasts about 3160ms (≈ 0.0009 h) at about 100mA, namely ≈ 0.09 mAh. We assume the duration of the sleep period is X hours at 10 μ A, namely 0.01 X mAh. The whole consumption for a cycle is $0.03mAh + 0.09mAh + 0.01XmAh$. The ESP32 is powered by a common 3.6v, 1.5Ah LiFePo4 battery. This guarantees a number of cycles $1500mAh / (0.12 + 0.01X)mAh$. The length of a cycle is $0.0016 + X$ hours. The whole duration in hours will be $1500 / (0.12 + 0.01X) \cdot (0.0016 + X)$ and with $X \geq 0.73$ hours, we meet requirement **R3**, namely a duration of 1 year.

Note that the smart contract gets in input an unsigned int made of 32 bytes, so we can pack up to 32 readings (they need 1 byte), into a single call, thus reducing the overall number of required transmissions.

IV. CONCLUSIONS

A comprehensive solution for seed preservation should address all the steps of the orthodox seed conservation pipeline in seed genebanks [8]. While this paper focuses exclusively

⁷<https://www.infura.io/>

⁸<https://www.ti.com/lit/ds/symlink/ina219.pdf>

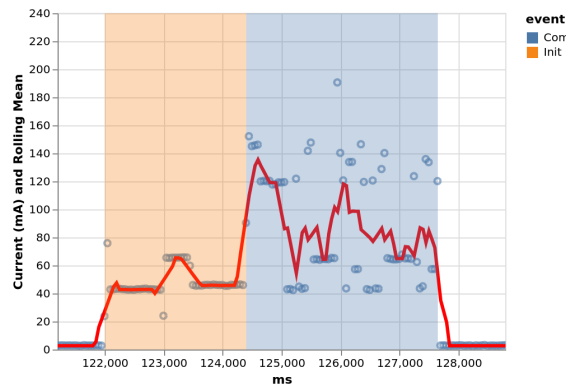


Fig. 2. Sampling current consumption with the INA219

on the storage phase, this step represents a critical component of the conservation process, as it directly affects the long-term maintenance of seed viability.

Our analysis showed that even SGSV, one of the world's most renowned backup seed facilities, exhibits limited redundancy for certain seed samples. This observation motivated our vision of Collective Seed Storage (CSS), a novel shared-economy approach that leverages domestic freezers as a distributed infrastructure for long-term seed storage. The proposed CSS framework combines IoT-based monitoring of storage conditions with blockchain-based incentive mechanisms to support user participation. Our proof of concept demonstrates the technical feasibility of the main components of this approach, showing that acceptable subzero storage conditions can be achieved in domestic freezers.

V. FUTURE WORK

Several research directions remain open. First, although our PoC confirms the feasibility of achieving subzero conditions in domestic freezers, further investigation is required to assess how closely these conditions can approach the ideal storage requirements prescribed in Standard 4.2.3 [7]. In particular, long-term storage experiments and the analysis of variability across different freezer models are necessary to evaluate the robustness of the proposed approach. An interesting research direction consists of investigating energy harvesting techniques that exploit the thermal gradient between the inside and outside of the freezer to enable energy-neutral operation.

From a socio-technical perspective, we aim to conduct a user study to investigate users' willingness to participate and explore more sophisticated engagement mechanisms, potentially informed by game-theoretical frameworks. In parallel, we plan to investigate the integration of CSS into the Multilateral System of the International Treaty on Plant Genetic Resources for Food and Agriculture [2], including the governance of seed exchanges among distributed participants. Beyond technical feasibility, future research should evaluate economic costs and develop strategies capable of fostering long-term participation.

VI. BROADER IMPACT AND SDGs

The CSS framework contributes to several United Nations Sustainable Development Goals (SDGs). It primarily supports SDG 2 (Zero Hunger) by addressing the risks associated with genetic erosion and by providing a distributed backup infrastructure for safeguarding crop diversity. By repurposing existing domestic freezer capacity through a shared-economy model, CSS also aligns with SDG 12 (Responsible Consumption and Production), promoting efficient use of existing resources rather than requiring additional energy-intensive infrastructure. Furthermore, by preserving the genetic resources needed for crop adaptation to climate-related challenges, CSS contributes to SDG 13 (Climate Action). Finally, its potential integration with the Multilateral System and the International Treaty on Plant Genetic Resources for Food and Agriculture supports SDG 17 (Partnerships for the Goals) by fostering international cooperation in the preservation and exchange of plant genetic resources.

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